

DIFFERENTIABILITY



SYNOPSIS

Differentiability at a point :

- (i) A function $f(x)$ is differentiable at a point

$x = a$, if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists finitely and it is denoted by $f'(a)$

$$\text{i.e. } f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

(ii) The right hand derivative of $f(x)$ at $x = a$ is denoted by $f'(a+)$ and is defined as

$$\text{i.e. } f'(a+) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

(iii) The left hand derivative of $f(x)$ at $x = a$ is denoted by $f'(a-)$ and is defined as

$$\text{i.e. } f'(a-) = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h}$$

W.E.-1 : The differentiability of

$$f(x) = \begin{cases} x \left(\frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} \right), & x \neq 0 \\ 0, & x = 0 \end{cases} \quad \text{at } x = 0$$

Sol: We have $f'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$

$$= \lim_{h \rightarrow 0} \left(\frac{e^{-2/h} - 1}{e^{-2/h} + 1} \right) = \frac{0 - 1}{0 + 1} = -1$$

$$\text{Similarly } f'(0+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = 1$$

$\therefore \text{LHD} \neq \text{RHD}$

$\therefore f$ is not differentiable at $x = 0$

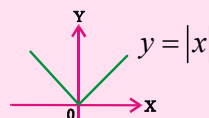
(iv) If f is differentiable at $x = a$ then f is also continuous at $x = a$. However the converse need not be true.

(v) If f is not continuous at $x = a$ then f is not differentiable at $x = a$

W.E.-2: $f(x) = \sin x$ is differentiable at $x = 0$
 $\Rightarrow f$ is continuous at $x = 0$

W.E.-3: Examine the continuity and differentiability of $f(x) = |x|$ at $x = 0$

Sol:



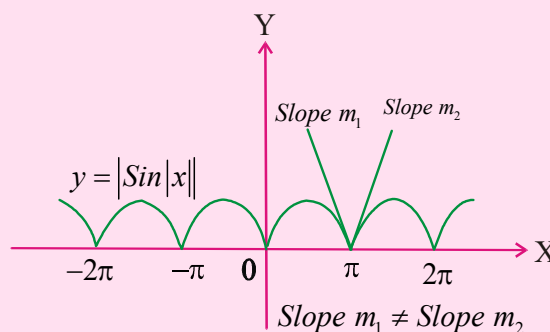
Hence $y = |x|$ is continuous everywhere but not differentiable at $x = 0$
 $(\because \text{sharp corner at } x = 0)$

W.E.-4: Examine the continuity and differentiability of $y = |\sin|x||$ at $x = n\pi, n \in I$

Sol: It is clear from the graph that $y = |\sin|x||$ is continuous everywhere but not differentiable at $x = \dots -2\pi, -\pi, 0, \pi, 2\pi, \dots$

i.e., $x = n\pi, n \in I$

We observe that at all integral values of π , f has a sharp corner



Differentiability of a function over an interval :

- i) A function $f(x)$ defined on an (a, b) is said to be differentiable in (a, b) if it is differentiable at each point of (a, b)
- ii) A function $f(x)$ defined on $[a, b]$ is said to be differentiable or derivable if
- f is differentiable from the right at a .
 - f is differentiable at every point on (a, b)

- c) f is differentiable from the left at b .
- iii) A function f is said to be a differentiable function, if it is differentiable at every point on its domain.
- iv) Exponential, logarithmic, trigonometric, inverse trigonometric functions are differentiable in their domain.
- v) Polynomial, constant functions are differentiable at each point ' x ', where $x \in \mathbb{R}$

Standard Results

$f(x)$	$g(x)$	$f(x) \pm g(x)$	$f(x).g(x)$
Differentiable	Differentiable	Differentiable	Differentiable
Differentiable	Non Differentiable	Non Differentiable	May be or not
Non Differentiable	Non Differentiable	May be or not	May be or not

- i) $|x - a|$ is not differentiable at $x = a$
- ii) $(x - a)^n |x - a|$ is differentiable when $n \geq 1$ and is not differentiable when $n < 1$
- iii) $\text{Sgn}(x - a)$ is not differentiable at $x = a$
- iv) $x^n \sin \frac{1}{x}, x^n \cos \frac{1}{x}$ are differentiable when $n > 1$ and are not differentiable when $n \leq 1$
- v) $\{x\}, [x]$ are not differentiable at all integral points of x
- vi) $\frac{d}{dx} \int_{\alpha(x)}^{\beta(x)} f(t) dt = f(\beta(x)) \frac{d}{dx} \beta(x) - f(\alpha(x)) \frac{d}{dx} \alpha(x)$

W.E.-5 : Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable function &

$$f(1) = 4 \text{ then } g(x) = \lim_{t \rightarrow 1} \int_4^{f(x)} \frac{2t}{x-1} dt =$$

$$\text{Sol: } g(x) = \lim_{x \rightarrow 1} \int_4^{f(x)} \frac{2t}{x-1} dt = \lim_{x \rightarrow 1} \frac{\int_4^{f(x)} 2t dt}{x-1}$$

Apply L-Hospital rule

$$= 2 \lim_{x \rightarrow 1} \frac{f(x) f'(x) - 4 \cdot 0}{1}$$

$$= \lim_{x \rightarrow 1} 2f(x) f'(x) = 2f(1) f'(1) = 8f'(1)$$

Differentiability of Functional Equations

- (i) if $f(x+y) = f(x).f(y)$ then

$$f'(x) = f'(0).f(x)$$

$$\text{Proof: } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x)(f(h)-1)}{h}$$

$$= f(x).f'(0)$$

- (ii) Functional equation relations.

a) $f(x+y) = f(x).f(y) \forall x, y$

$$\Rightarrow f(x) = a^x (a > 0)$$

b) $f(x+y) = f(x) + f(y) \forall x, y \in \mathbb{R}$

$$\Rightarrow f(x) = kx$$

c) $f(xy) = f(x).f(y) \forall x, y \in \mathbb{R}$

$$\Rightarrow f(x) = x^n$$

d) $f(xy) = f(x) + f(y) \forall x, y \in \mathbb{R}^+$

$$\Rightarrow f(x) = k \log x (x > 0)$$

e) $f(x) f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \forall x \in \mathbb{R} - \{0\}$

$$\Rightarrow f(x) = 1 \pm x^n$$

(f) $f\left(\frac{mx+ny}{m+n}\right) = \frac{mf(x)+nf(y)}{m+n}, m+n \neq 0$

$$\Rightarrow f(x) = ax + b$$

W.E-6: If $f(x+y) = f(x).f(y) \forall x, y \in \mathbb{R}, f(5) = 2,$

$$f'(0) = 3 \text{ then } f'(5) = \quad [\text{AIEEE 2002}]$$

Sol: $f'(x) = f'(0).f(x)$

$$f'(5) = f'(0).f(5) = (3)(2) = 6$$



C.U.Q.

- If both $f(x)$ & $g(x)$ are differentiable functions at $x = x_0$, then the function defined as, $h(x) = \text{Maximum}\{f(x), g(x)\}$:
 - is always differentiable at $x = x_0$
 - is never differentiable at $x = x_0$
 - is differentiable at $x = x_0$ when $f(x_0) \neq g(x_0)$
 - can not be differentiable at $x = x_0$ if $f(x_0) = g(x_0)$.
- If $f(x)$ differentiable everywhere then
 - $|f(x)|$ is differentiable everywhere
 - $|f|^2$ is differentiable everywhere
 - $[f(x)]$ is differentiable everywhere
 - $\text{Sgn}(f(x))$ is differentiable everywhere
- If $f(x) = (x - a)g(x)$ and $g(x)$ is continuous at $x = a$, then $f'(a) =$
 - $a g(a)$
 - $g'(a)$
 - $g(a)$
 - $ag'(a)$
- If $f(x) = \text{sgn}(x)$ then $f'(0) =$
 - 0
 - 1
 - 1
 - does not exist
- If $f(0) = 0$, $f'(0) = 2$ then derivative of $y = f(f(x))$ at $x = 0$ is
 - 1
 - 2
 - 4
 - 8

C.U.Q - KEY

- 3
- 2
- 3
- 4
- 3

C.U.Q - HINTS

- Consider the graph of $h(x) = \max(x, x^2)$ at $x = 0$ and $x = 1$
for Differentiability: $h(x) = \max(x^2, -x^2)$
- Conceptual.
- By using $f(a) = 0$, we get $f'(a) = g(a)$
- $f(x) = \begin{cases} 0, & x = 0 \\ -1, & x < 0 \\ 1, & x > 0 \end{cases}$ $f(x)$ is not continuous at $x=0$
- $\frac{dy}{dx} = f'(f(x)) \cdot f'(x)$



LEVEL-I(C.W.)

DIFFERENTIABILITY AT A POINT

- Let $f(x) = x^{3/2} - \sqrt{x^3 + x^2}$ then
 - LHD at $x=0$ exist but RHD at $x=0$ does not exist
 - $f(x)$ is not differentiable at $x=0$
 - RHD at $x=0$ exist but LHD at $x=0$ does not exist
 - $f(x)$ is differentiable and continuous at $x=0$
- If $f(x) = x \cdot \left(\frac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}} \right)$, $x \neq 0$ ($a > 0$), $f(0) = 0$
then
 - f is differentiable at $x=0$
 - f is not differentiable at $x=0$
 - f is not continuous at $x=0$
 - $\lim_{x \rightarrow 0} f(x)$ does not exist
- Let $f(x) = \begin{cases} x^2, & \text{if } x \leq x_0 \\ ax + b, & \text{if } x > x_0 \end{cases}$. If f is differentiable at x_0 then
 - $a = x_0, b = -x_0$
 - $a = 2x_0, b = -x_0^2$
 - $a = 2x_0, b = x_0^2$
 - $a = x_0, b = x_0^2$
- Let $f(x) = |x-a|\phi(x)$, where ϕ is a continuous function and $\phi(a) \neq 0$. Then
 - $f'(a+) = \phi'(a)$
 - f is differentiable at $x=a$
 - $f'(a+) = -\phi'(a)$
 - $f'(a-) = -\phi(a)$
- The left-hand derivative of $f(x) = [x] \sin \pi x$ at $x = k$, k is an integer is
 - $(-1)^k(k-1)\pi$
 - $(-1)^{k-1}(k-1)\pi$
 - $(-1)^k k \pi$
 - $(-1)^{k-1} k \pi$
- If $f(x) = \begin{cases} 1 & x < 0 \\ x^2 & x \geq 0 \end{cases}$ then at $x=0$
 - f' is differentialble
 - f is continuous
 - f is differentiable
 - f is not continuous

7. If $f(x) = |x - a| + |x + b|$, $x \in \mathbb{R}, b > a > 0$. Then

- 1) $f'(a+) = 1$ 2) $f'(a+) = 0$
 3) $f'(-b+) = 0$ 4) $f'(-b+) = 1$

DIFFERENTIABILITY OVER AN INTERVAL

8. Let $f(x) = \begin{cases} x & x < 1 \\ 2 - x & 1 \leq x \leq 2 \\ -2 + 3x - x^2 & x > 2 \end{cases}$ then $f(x)$ is

- 1) differentiable at $x=1$
 2) differentiable at $x=2$
 3) differentiable at $x=1$ and $x=2$
 4) not differentiable at $x=0$

9. Let $f(x) = \begin{cases} x^2/2 & 0 < x \leq 1 \\ 2x^2 - 3x + 3 & 1 < x < 2 \end{cases}$ then which is

correct

- 1) f is continuous in $[0, 2]$
 2) f' is continuous in $[0, 2]$
 3) $f(x)$ is discontinuous at $x=1$
 4) $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$

10. Let $f(x) = a \sin|x| + b e^{|x|}$ is differentiable when

- 1) $a = -b$ 2) $a = b$ 3) $a = 0$ 4) $b = 0$

LEVEL - I (C.W)-KEY

- 1) 3 2) 2 3) 2 4) 4 5) 1 6) 4
 7) 3 8) 2 9) 3 10) 1

LEVEL - I (C.W)-HINTS

1. $f(x) = x^{3/2} - \sqrt{x^2 + x^2}$
 $\therefore$ L.H.D at $x=0$ does not exist as $D_f = [0, \infty)$

2. $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x(a^{1/x} - a^{-1/x})}{a^{1/x} + a^{-1/x}} = \lim_{x \rightarrow 0^+} \frac{x(1 - a^{-2/x})}{1 + a^{-2/x}} = 0$

$$\text{also, } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x \left(\frac{a^{2/x} - 1}{a^{2/x} + 1} \right) = 0$$

so f is continuous at $x=0$

$$f'(0+) = \lim_{x \rightarrow 0^+} \frac{h(a^{1/h} - a^{-1/h})}{h(a^{1/h} + a^{-1/h})} = \lim_{x \rightarrow 0^+} \frac{1 - a^{-2/h}}{1 + a^{-2/h}} = 1$$

similarly $f'(0-) = -1$ hence, f is not differentiable at $x=0$

3. Since f is differentiable so it is continuous also,

$$= \lim_{h \rightarrow 0} \frac{a+b-h+h-a-b}{h} = 0$$

therefore $x^2_0 = f(x_0) = \lim_{x \rightarrow x_0} f(x) = ax_0 + b$

$$\text{also, } \lim_{h \rightarrow 0^+} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{a(x_0+h) + b - x^2_0}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{x^2_0 + ah - x^2_0}{h} = a (\because x^2_0 = ax_0 + b)$$

$$\text{thus } a = f'(x_0-) = \lim_{h \rightarrow 0^+} \frac{(x_0+h)^2 - x^2_0}{h} = 2x_0$$

hence $x^2_0 = 2x^2_0 + b$ then $b = -x^2_0$

$$4. \quad f'(a+) = \lim_{h \rightarrow 0^+} \frac{|a+h-a|\phi(a+h)-0}{h} \\ = \lim_{h \rightarrow 0^+} \frac{|h|\phi(a+h)}{h} = \lim_{h \rightarrow 0^+} \phi(a+h) = \phi(a)$$

similarly $f'(a-) = -\phi(a)$

5. Clearly $f(k)=0$, so the left hand derivative is equal

$$\text{to } \lim_{h \rightarrow 0^-} \frac{f(k+h) - f(k)}{h} \\ = \lim_{h \rightarrow 0^-} \frac{[k+h]\sin(k+h)\pi}{h} = \lim_{h \rightarrow 0^-} \frac{(k-1)\sin(k\pi+h\pi)}{h} \\ = \lim_{h \rightarrow 0^-} \frac{(k-1)(-1)^k \sinh \pi}{h} \quad (\text{since } h < 0) = (k-1)(-1)^k \pi$$

$$6. \quad f(0^-) \neq f(0^+)$$

$$7. \quad f'(x) = \begin{cases} -2, & \text{if } x < -b \\ 0, & \text{if } -b \leq x \leq a \\ 2, & \text{if } x > a \end{cases}; \quad f'(a+) = 2, f'(-b+) = 0$$

$$8. \quad f'(x) = \begin{cases} 1, & \text{if } x < 1 \\ -1, & \text{if } 1 \leq x \leq 2 \\ 3-2x, & \text{if } x > 2 \end{cases}$$

$$f'(1-) = 1, f'(1+) = -1, f'(2-) = -1, f'(2+) = -1$$

$$f'(2-) = f'(2+)$$

$$9. \quad \lim_{x \rightarrow 1^-} f(x) = 1/2; \quad \lim_{x \rightarrow 1^+} f(x) = 2$$

$$10. \quad f(x) = \begin{cases} -a \sin x + b e^{-x} & \text{if } x < 0 \\ a \sin x + b e^x & \text{if } x > 0 \end{cases}; \quad f'(x) = \begin{cases} -a \cos x - b e^{-x} & \text{if } x < 0 \\ a \cos x + b e^x & \text{if } x > 0 \end{cases}$$

then $f(0-) = -a - b$ and $f(0+) = a + b$

If $a = -b$, then $f(0-) = f(0+)$



LEVEL-I(H.W.)

DIFFERENTIABILITY AT A POINT

1. Let $f(x) = |x-1| + |x+1|$

- 1) $f(x)$ is differentiable at $x = \pm 1$
 2) $f(x)$ is not differentiable at $x = \pm 1$
 3) $f(x)$ is neither continuous nor differentiable at $x = \pm 1$
 4) $f(x)$ is not continuous at $x=0$

2. Let $f(x) = \begin{cases} 0 & \forall x < 0 \\ x^2 & \forall x \geq 0 \end{cases}$ then $\forall x \in \mathbb{R}$
- 1) f is differentiable
 - 2) f, f' both are continuous
 - 3) f is differentiable, f' is continuous
 - 4) All of these
3. Let $f(x) = \begin{cases} \frac{x}{1+2^{\frac{1}{x}}}, & x \neq 0 \\ 0 & x = 0 \end{cases}$ then
- 1) LHD $f(x)$ at $x=0$ is 1
 - 2) RHD of $f(x)$ at $x=0$ is not equal to zero
 - 3) $f(x)$ is differentiable at $x=0$
 - 4) $\lim_{x \rightarrow 0} f'(x) = 1$
4. If $f(x) = |\log_{10} x|$ then at $x=1$
- 1) f is not continuous
 - 2) f is continuous but not differentiable
 - 3) f is differentiable
 - 4) the derivative is 1
5. $f(x) = \begin{cases} |x-4| & \text{for } x \geq 1 \\ x^{\frac{3}{2}} - x^2 + 3x + \frac{1}{2} & \text{for } x < 1 \end{cases}$, then
- 1) $f(x)$ is continuous at $x=1$ and $x=4$
 - 2) $f(x)$ is differentiable at $x=4$
 - 3) $f(x)$ is continuous and differentiable at $x=1$
 - 4) $f(x)$ is only continuous at $x=1$
6. If $f'(3) = 2$, then $\lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{2h^2}$ is
- 1) 1
 - 2) 2
 - 3) 3
 - 4) $1/2$
7. If $[\cdot]$ denote the greatest integer function and $f(x) = [\tan^2 x]$, then
- 1) $\lim_{x \rightarrow 0} f(x)$ does not exist
 - 2) f is not continuous at $x=0$
 - 3) $f(x)$ is differentiable at $x=0$
 - 4) $f'(0) = 1$
8. If $f(x) = |x|e^x$, then at $x=0$
- 1) f is continuous
 - 2) f is continuous but not differentiable
 - 3) f is differentiable
 - 4) the derivative is 1

DIFFERENTIABILITY OVER AN INTERVAL

9. The set of all points where the function $f(x) = \sqrt{1 - e^{-x^2}}$ is differentiable is
- 1) $(0, \infty)$
 - 2) $(-\infty, \infty)$
 - 3) $(-\infty, \infty) - \{0\}$
 - 4) $(-\infty, \infty) - \{0, 1, 2\}$
10. The set of all points where $f(x) = 2x|x|$ is differentiable
- 1) $(-\infty, \infty)$
 - 2) $(-\infty, \infty) - \{0\}$
 - 3) $(0, \infty)$
 - 4) $[0, \infty)$

LEVEL-I (H.W)-KEY

- 1) 2 2) 4 3) 1 4) 2 5) 1
6) 2 7) 3 8) 2 9) 3 10) 1

LEVEL-I (H.W)-HINTS

1. $f(x) = \begin{cases} -2x, & \text{if } x < -1 \\ 2, & \text{if } -1 \leq x < 1 \\ 2x, & \text{if } x \geq 1 \end{cases} \Rightarrow f'(x) = \begin{cases} -2, & \text{if } x < -1 \\ 0, & \text{if } -1 \leq x < 1 \\ 2, & \text{if } x \geq 1 \end{cases}$
- $f'(-1^-) = -2, f'(-1^+) = 0, f'(1^-) = 0, f'(1^+) = 2$
 $\therefore f(x)$ is not differentiable at $x = \pm 1$
2. $f(x) = \begin{cases} 0 & \forall x < 0 \\ x^2 & \forall x \geq 0 \end{cases} \Rightarrow f'(x) = \begin{cases} 0 & \forall x < 0 \\ 2x & \forall x \geq 0 \end{cases}$
- then $f'(x) = \begin{cases} 0 & \forall x < 0 \\ 2 & \forall x \geq 0 \end{cases}$
- $f'(0^-) = f'(0^+) = 0$ and
 $f''(0^-) = 0, f''(0^+) = 2$
 $\therefore f$ and f' are differentiable
3. $f'(0^-) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{\frac{x}{1+2^{\frac{1}{x}}} - 0}{x} = \frac{1}{1+2^{\frac{1}{0^-}}} = \frac{1}{1+2^{-\infty}} = 1$
- $f'(0^+) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{\frac{x}{1+2^{\frac{1}{x}}} - 0}{x} = \frac{1}{1+2^{\frac{1}{0^+}}} = \frac{1}{1+2^{\infty}} = 0$
 $\therefore$ L.H.D of $f(x)$ at $x=0$ is 1
4. Since $g(x) = |x|$ is continuous but not differentiable at $x=0$, so f is continuous but not differentiable for $\log_{10} x = 0$ i.e for $x=1$

5. Since $g(x)=|x|$ is a continuous function and $\lim_{x \rightarrow 1+} f(x) = 3 = \lim_{x \rightarrow 1-} f(x)$, so f is continuous function. In particular f is continuous at $x=1$ and $x=4$ f is clearly not differentiable at $x=4$ Since $g(x)=|x|$ is not differentiable at $x=0$. Now

$$f'(1+) = \lim_{h \rightarrow 0+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0+} \frac{|-3+h| - 3}{h} = -1$$

$$f'(1-) = \lim_{h \rightarrow 0-} \frac{(\frac{1}{2})(1+h)^3 - (1+h)^2 + 3(1+h) + (\frac{1}{2}) - 3}{h}$$

$$= \lim_{h \rightarrow 0-} \frac{(\frac{1}{2})(h^3 + 3h^2 + 3h) - (h^2 + 2h) + 3h}{h} = \frac{5}{2}$$

$$6. \lim_{h \rightarrow 0} \left(\frac{f(3+h^2) - f(3-h^2)}{2h^2} \right)$$

$$= \frac{1}{2} \left\{ \lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3)}{h^2} - \lim_{h \rightarrow 0} \frac{f(3-h^2) - f(3)}{h^2} \right\}$$

$$= \frac{1}{2} \{ f'(3) - (-f'(3)) \} = \frac{1}{2} \times 2f'(3) = 2$$

7. $0 < x < \pi/4$, $[\tan^2 x] = 0$. Also $\tan^2 x$ is an even function

$$\lim_{x \rightarrow 0+} \left(\frac{f(0+x) - f(0)}{x} \right) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = 0$$

$\therefore f$ is continuous at $x=0$ and differentiable $x=0$. Also $f'(0)=0$

8. $L.H.D = \lim_{x \rightarrow 0-} \frac{-xe^x}{x} = -1$

$$R.H.D = \lim_{x \rightarrow 0+} \frac{xe^x}{x} = 1$$

9. $f'(x) = \frac{xe^{-x^2}}{\sqrt{1-e^{-x^2}}}$ is not differentiable only at $x=0$

10. $f(x) = \begin{cases} 2x^2 & x \geq 0 \\ -2x^2 & x < 0 \end{cases}$ is differentiable everywhere.



LEVEL-II(C.W.)

DIFFERENTIABILITY AT A POINT

1. If $f(x) = \begin{cases} -\frac{1}{2}x^2, & \text{for } x < 1 \\ \frac{3}{2}x^2 + 1, & \text{for } x \geq 1 \end{cases}$, then

1) f is differentiable everywhere on $\mathbb{R}$

2) $f'(1-) = -1$ and $f'(1+) = 0$

3) $f'(1-) = -1$ and $f'(1+) = 3$

4) $f'(1-) = 1$ and $f'(1+) = -1$

2. If $f(x) = \frac{x\sqrt{x}}{x+2}$ then derivative of $f(x)$ at

$x=0$ is

1) 0 2) 1 3) $1/2$ 4) does not exist

3. Let f and g be differentiable functions satisfying $g(a)=b, g'(a)=2$ and $f \circ g = I$ (identity function). Then $f'(b)$ is equal to

1) $1/2$ 2) 2 3) $2/3$ 4) 3

4. Let $f(x) = x^2|x|$ then the set of values where $f(x)$ is three times differentiable is

1) Infinite 2) 2 3) 3 4) 0

5. If $f(x) = p|\sin x| + qe^{|x|} + r|x|^3$ and $f(x)$ is differentiable at $x=0$, then

1) $q+r=0$; p is any real number

2) $p+q=0$; r is any real number

3) $q=0, r=0$; p is any real number

4) $r=0, p=0$; q is any real number

6. If $f(x) = \begin{cases} -3x+2, & x < 1 \\ \frac{1}{2}x^2+7, & x \geq 1 \end{cases}$, then which of the

following is not true

1) $f'(1+) = 1$ 2) $f'(1-) = -3$

3) $f'(1-) = f'(1+) = 1$

4) f is not differentiable at $x=1$

7. Let $f(x) = \frac{\sin 4\pi[x]}{1+[x]^2}$, where $[x]$ is the greatest

integer less than or equal to x , then

1) $f(x)$ is not differentiable at some points

2) $f(x)$ exists but is different from zero

3) LHD (at $x=0$) = 0, RHD (at $x=1$) = 0

4) $f(x)=0$ but f is not a constant function

8. The function given by $y = ||x| - 1|$ is differentiable for all real numbers except the points
 1) $\{0, 1, -1\}$ 2) ± 1 3) 1 4) -1
9. If $f : R \rightarrow R$ is an even function which is twice differentiable on R and $f''(\pi) = 1$ then $f''(-\pi)$ is
 1) -1 2) 0 3) 1 4) 2

DIFFERENTIABILITY OVER AN INTERVAL

10. Let a function $y = f(x)$ be defined as $x = 2t - |t|$, $y = t^2 + t|t|$, Where $t \in R$ then $f(x)$ is
 1) Continuous and differentiable in $[-1, 1]$
 2) Continuous but not differentiable in $[-1, 1]$
 3) Continuous $[-1, 1]$ and differentiable in $(-1, 1)$ only
 4) Discontinuous on $[-1, 1]$
11. Let $f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 < x \leq 2 \end{cases}$ and $g(x) = |f(x)| + f(x)$ then the number of points which $g(x)$ is non differentiable, is
 1) at most one point 2) 2
 3) exactly one point 4) infinite
12. If $f(x) = \begin{cases} [\cos \pi x], & x < 1 \\ |x - 2|, & 1 \leq x < 2 \end{cases}$, then $f(x)$ is
 1) discontinuous and non-differentiable at $x = -1$ and $x = 1$
 2) continuous and differentiable at $x = 0$
 3) not differentiable at $x = 1/2$
 4) continuous but not differentiable at $x = 0$

DIFFERENTIABILITY OF FUNCTIONAL EQUATIONS

13. Let $f(x+y) = f(x)f(y)$ and $f(x) = 1 + xg(x)G(x)$, where $\lim_{x \rightarrow 0} g(x) = a$ and $\lim_{x \rightarrow 0} G(x) = b$. Then $f'(x)$ is equal to
 1) $1 + ab$ 2) ab 3) $f(x)$ 4) $abf(x)$

14. If $f : R \rightarrow R$ be a differentiable function, such that $f(x+2y) = f(x) + f(2y) + 4xy$ for all $x, y \in R$ then
 1) $f'(1) = f'(0) + 1$ 2) $f'(1) = f'(0) - 1$
 3) $f'(0) = f'(1) + 2$ 4) $f'(0) = f'(1) - 2$
15. Let f be a differentiable function satisfying the condition $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$, for all $x, y (\neq 0) \in R$ and $f(y) \neq 0$. If $f'(1) = 2$, then $f'(x)$ is equal to
 1) $2f(x)$ 2) $\frac{f(x)}{x}$ 3) $2xf'(x)$ 4) $\frac{2f(x)}{x}$

LEVEL-II (C.W)-KEY

- 1) 3 2) 1 3) 1 4) 1 5) 2 6) 3
 7) 3 8) 1 9) 3 10) 1 11) 3 12) 3
 13) 4 14) 4 15) 4

LEVEL-II (C.W)-HINTS

1. $f'(x) = \begin{cases} -x & \text{if } x < 1 \\ 3x & \text{if } x \geq 1 \end{cases}$
2. $f'(0^+) = \lim_{x \rightarrow 0^+} \left(\frac{f(x) - f(0)}{x - 0} \right) = \lim_{x \rightarrow 0^+} \left(\frac{\frac{x\sqrt{x}}{x+2} - 0}{x - 0} \right)$
3. $\text{fog} = I \Rightarrow \text{fog}(x) = x$ for all $x \Rightarrow f'(g(x))g'(x) = 1$ for all x
 $\Rightarrow f'(g(a)) = \frac{1}{g'(a)} = \frac{1}{2} \Rightarrow f'(b) = \frac{1}{2} (\because g(a) = b)$
4. $f(x) = \begin{cases} -x^3, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ x^3, & \text{if } x > 0 \end{cases}$ and $f''(x) = \begin{cases} -6, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ 6, & \text{if } x > 0 \end{cases}$
 $f''(0^-) \neq f''(0^+)$ $f''(x)$ is exists except at $x = 0$
 $\therefore$ number of points = ∞
5. For $-\frac{\pi}{2} < x \leq 0$, $f(x) = -p \sin x + qe^{-x} - rx^3$, so
 $= -p - q$

For $0 < x < \frac{\pi}{2}$, $f(x) = p \sin x + qe^x + rx^3$,

$$f'(0+) = \lim_{x \rightarrow 0+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0+} \left[\frac{p \sin x}{x} + q \left(\frac{e^x - 1}{x} \right) - rx^2 \right] = p + q$$

For f to be differentiable at $x=0$, we must have

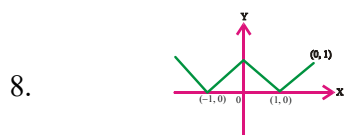
$$p+q = -p - q \Rightarrow p+q = 0.$$

$$6. \quad f'(x) = \begin{cases} -3, & \text{if } x < 1 \\ x, & \text{if } x \geq 1 \end{cases}$$

$$7. \quad \text{We have } \frac{\sin 4\pi[x]}{1+[x]^2} = 0, \quad \forall x$$

$[\cdot] : 4\pi[x]$ is an integral multiple of ' π '

$$\Rightarrow f(x) = 0 \text{ for all } x$$



9. $f : R \rightarrow R$ is an even function

$f'(x)$ is an odd function

Hence $f''(x)$ is even

$$f''(-\pi) = f''(\pi) = 1$$

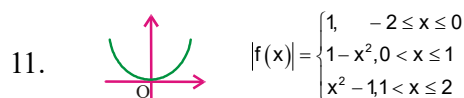
10. When $t \geq 0$,

we have $x = 2t - t$ and

$$y = t^2 + t^2 = 2t^2 \Rightarrow y = 2x^2, x \geq 0$$

When $t < 0$, we have

$$x = 2t + t = 3t \text{ and } y = t^2 - t^2 = 0 \Rightarrow y = 0 \text{ for all } x < 0$$



$$\text{and } f(x) = x^2 - 1, -2 \leq x \leq 2$$

$$\therefore g(x) = \begin{cases} x^2, & -2 \leq x \leq 0 \\ 0, & 0 < x \leq 1 \\ 2(x^2 - 1), & 1 < x \leq 2 \end{cases}$$

(by adding the function in proper domains)

$\therefore g(x)$ is differentiable everywhere except at $x=1$

$$12. \quad \text{We have, } f(x) = \begin{cases} [\cos \pi x] & x < 1 \\ |x-2| & 1 < x < 2 \end{cases}$$

$$= \begin{cases} 2-x, & 1 \leq x < 2 \\ -1, & 1/2 < x < 1 \\ 0, & 0 < x \leq 1/2 \\ 1, & x = 0 \\ 0, & -1/2 \leq x < 0 \\ -1, & -1/2 < x < -1/2 \end{cases}$$

It is evident from the definition that $f(x)$ is discontinuous at $x=1/2$

$$13. \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$[\because f(x+y) = f(x)f(y)]$$

$$= f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = f(x) \lim_{h \rightarrow 0} \frac{1 + hg(h)G(h) - 1}{h}$$

$$= f(x) \lim_{h \rightarrow 0} g(h)G(h) = f(x) \lim_{h \rightarrow 0} G(h) \lim_{h \rightarrow 0} g(h) = abf(x)$$

$$14. \quad f(x+2y) = f(x) + f(2y) + 4xy \text{ for all } x, y \in R$$

putting $x = y = 0$, we get $f(0) = 0$

$$\text{Now, } f(x+2y) = f(x) + f(2y) + 4xy$$

$$\Rightarrow \frac{f(x+2y) - f(x)}{2y} = 2x + \frac{f(2y)}{2y}$$

$$\Rightarrow \lim_{y \rightarrow 0} \frac{f(x+2y) - f(x)}{2y}$$

$$= \lim_{y \rightarrow 0} \left\{ 2x + \frac{f(2y) - f(0)}{2y} \right\}$$

$$\Rightarrow f'(x) = 2x + f'(0) \text{ for all } x$$

$$\Rightarrow f'(1) = 2 + f'(0)$$

$$15. \quad f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}, \text{ replacing } x \text{ and } y \text{ both by } 1,$$

we get

$$f(1) = \frac{f(1)}{f(1)} \Rightarrow f(1) = 1$$

$$\text{Now } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow f'(x) = f(x) \lim_{h \rightarrow 0} \left\{ \frac{\frac{f(x+h)}{f(x)} - 1}{h} \right\}$$

$$\Rightarrow f'(x) = f(x) \lim_{h \rightarrow 0} \left\{ \frac{f\left(\frac{x+h}{x}\right) - 1}{\frac{h}{x}} \right\}$$

$$\Rightarrow f'(x) = \frac{f(x)}{x} \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right) - f(1)}{\frac{h}{x}}$$

$$\Rightarrow f'(x) = \frac{f(x)}{x} f'(1) = \frac{2f(x)}{x}$$



LEVEL-II(H.W.)

DIFFERENTIABILITY AT A POINT

1. If $f(x) = |x| + |\sin x|$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then its left hand derivative at $x = 0$ is (Eam-2011)
1) 0 2) -1 3) -2 4) -3

2. The function

$$f(x) = (x^2 - 1)|x^2 - 3x + 2| + \cos(|x|)$$

is not differentiable at

- 1) $x = -1$ 2) $x = 0$ 3) $x = 1$ 4) $x = 2$

3. Let $f(x) = \begin{cases} \frac{\sin|x^2 - 5x + 6|}{x^2 - 5x + 6}, & x \neq 2, 3 \\ 1, & x = 2 \text{ or } 3 \end{cases}$

the set of all points where f is differentiable is

- 1) $(-\infty, \infty)$ 2) $(-\infty, \infty) - \{2\}$
3) $(-\infty, \infty) - \{3\}$ 4) $(-\infty, \infty) - \{2, 3\}$

4. $f(x) = |\cos x|$ is not differentiable for the points given by $x =$

- 1) $\frac{\pi}{2}$ 2) $(2n+1)\pi, \forall n \in I$

- 3) $(2n+1)\frac{\pi}{2}, \forall n \in I$ 4) 0

5. Let 'f' be a function defined by $2f(\sin x) + f(\cos x) = x, \forall x$ then set of points where 'f' is not differentiable is

- 1) Set of all natural numbers
2) Set of all irrational numbers
3) $\{0, 1, -1\}$ 4) $\{1, -1\}$

6. If $f(x) = \begin{cases} (\cos x)^{\frac{1}{\sin x}} & \text{for } x \neq 0 \\ k & \text{for } x = 0 \end{cases}$ the value of k , so that f is differentiable at $x=0$ is

- 1) 0 2) 1 3) $\frac{1}{2}$ 4) 2

7. If $0 < a < 1$ and f is defined as

$$f(x) = \begin{cases} x^a & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases} \text{ then}$$

- 1) f is differentiable at $x=0$
2) f is not continuous
3) f is not differentiable at $x=0$
4) f is bounded

8. If $f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin x, & 0 \leq x < \frac{\pi}{2} \end{cases}$, then derivative of $f(x)$ at $x=0$

- 1) is equal to 1 2) is equal to 0
3) is equal to -1 4) does not exist

DIFFERENTIABILITY OVER AN INTERVAL

9. Let $h(x) = \min\{x, x^2\}$ for $x \in \mathbb{R}$. Then which of the following is correct

- 1) h is continuous for all x
2) h is differentiable for all x
3) $h'(x) = 1$ for all $x > 1$
4) h is not differentiable at 2 values of x

10. Let $f(x) = \begin{cases} 3^x & \forall |x| \leq 1 \\ 7 - x & \forall 1 < x < 7 \end{cases}$ then $f(x)$ is

- 1) continuous $\forall 1 \leq x \leq 7$ but not differentiable at $x=1$
2) continuous $-1 \leq x < 7$ & differentiable at $x=1$
3) neither continuous in $[-1, 7]$ nor differentiable at $x=1$
4) continuous & differentiable at $x=1$

DIFFERENTIABILITY OF FUNCTIONAL EQUATIONS

11. If $f(x+y) = 2f(x)f(y)$ all $x, y \in \mathbb{R}$ where

$f'(0) = 3$ and $f(4) = 2$, then $f'(4)$ is equal to

- 1) 6 2) 12 3) 4 4) 3

12. Let $f(x+y) = f(x)f(y)$ and $f(x) = 1 + (\sin 2x)g(x)$ where $g(x)$ is continuous. Then $f'(x)$ equals

- 1) $f(x)g(0)$ 2) $2f(x)g(0)$
3) $2g(0)$ 4) $2f(0)$

LEVEL-II (H.W)-KEY

- 1) 3 2) 4 3) 4 4) 3 5) 4 6) 2
7) 3 8) 4 9) 4 10) 3 11) 2 12) 2

LEVEL-II (H.W)-HINTS

- $x \rightarrow 0^-, f(x) = -x - \sin x$
- By verification $f'(2^-) \neq f'(2^+)$
 $\therefore f(x)$ is not differentiable at $x=2$
- The function is clearly differentiable except possible at $x=2, 3$
$$f'(2^+) = \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{\sin h(1-h) + h(1-h)}{h^2(-1+h)}$$
$$= -\lim_{h \rightarrow 0^+} \left(\frac{\sin h(1-h)}{h(1-h)} + 1 \right) \frac{1}{h} \text{ which does not exist}$$
- f is not differentiable at all points where $\cos x = 0$
- $2f(\sin x) + f(\cos x) = x \quad (1)$

Replacing 'x' by $\frac{\pi}{2} - x$,

$$2f(\cos x) + f(\sin x) = \frac{\pi}{2} - x \quad (2)$$

Solving (1) and (2) $3f(\sin x) = 3x - \frac{\pi}{2}$

(or) $f(\sin x) = x - \frac{\pi}{6}$

$\therefore f(x) = \sin^{-1} x - \frac{\pi}{6}$

$f'(x) = \frac{1}{\sqrt{1-x^2}}$ is not differentiable at

$x = 1, -1$

6. $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{\sin x}} = k$

$k = e^{\lim_{x \rightarrow 0} \left[\frac{1}{\sin x} \log(\cos x) \right]} = e^{\lim_{x \rightarrow 0} \frac{-\tan x}{\cos x}} = e^0 = 1$

7. $f'(0^+) = \lim_{x \rightarrow 0} \frac{x^a - 0}{x - 0} = \lim_{x \rightarrow 0} \frac{1}{x^{1-a}}$

$0 < a < 1 \Rightarrow (1-a) > 0 \Rightarrow f'(0) = \infty$, does not exist

8. We have (LHD at $x=0$) $= \left[\frac{d}{dx}(1) \right]_{x=0} = 0$

9.
$$h(x) = \begin{cases} x, & x \geq 1 \\ x^2, & 0 \leq x < 1 \\ x, & x < 0 \end{cases}$$

DIFFERENTIABILITY

From the graph it is clear that h is continuous. Also h is differentiable except possible at $x=0$ and 1

$$h'(x) = \begin{cases} 1, & x > 1 \\ 2x, & 0 < x < 1 \\ 1, & x < 0 \end{cases}$$

for $x=1$, $h'(1^+) = \lim_{t \rightarrow 1^+} \frac{h(1+t) - h(1)}{t} = \lim_{t \rightarrow 1^+} \frac{1+t-1}{t} = 1$

but $h'(1^-) = \lim_{t \rightarrow 1^-} \frac{h(1-t) - h(1)}{t} = \lim_{t \rightarrow 1^-} \frac{(1-t)^2 - 1}{t} = -2$

so h is not differentiable at 1

similarly $h'(0^+) = 0$ but $h'(0^-) = 1$

10. $f(x) = \begin{cases} 3^x & \forall |x| \leq 1 \\ 7-x & 1 < x < 7 \end{cases}$

$f'(1^-) = 3 \log 3$, $f'(1^+) = 0 - 1 = -1$, $f'(1^-) \neq f'(1^+)$

$\therefore f(x)$ is not differentiable at $x=1$

Hence f is not differentiable at $x=1$

11. $f'(4) = \lim_{h \rightarrow 0} \frac{f(4+h) - f(4)}{h}$

$\Rightarrow f'(4) = \lim_{h \rightarrow 0} \frac{f(4+h) - f(4+0)}{h}$

$\Rightarrow f'(4) = \lim_{h \rightarrow 0} \frac{2f(4)f(h) - 2f(4)f(0)}{h}$

$\Rightarrow f'(4) = \lim_{h \rightarrow 0} 2f(4) \left[\frac{f(h) - f(0)}{h - 0} \right]$

$f'(4) = 4 \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h - 0}$

$= 4f'(0) = 4 \times 3 = 12$

12. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h}$

$= f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = f(x) \lim_{h \rightarrow 0} \frac{1 + (\sin 2h)g(h) - 1}{h}$

$= f(x) \lim_{h \rightarrow 0} \frac{\sin 2h}{h} \cdot \lim_{h \rightarrow 0} g(h) = 2f(x)g(0)$



LEVEL-III

1. If $f(x) = |\cos x - \sin x|$, then $f'\left(\frac{\pi}{4}\right) =$

1) $\sqrt{2}$ 2) $-\sqrt{2}$ 3) 0 4) does not exist

2. Let $f(x)$ be a polynomial of degree two which is positive for all $x \in \mathbb{R}$.

$g(x) = f(x) + f'(x) + f''(x) + f'''(x) + x^2 f^{(iv)}(x)$,
then for any real x

1) $g(x) < 0$ 2) $g(x) > 0$ 3) $g(x) = 0$ 4) $g(x) \geq 0$

3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ & $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $g(x) = xf(x)$ then
 1) g is a differentiable function
 2) g is differentiable at 0 if f is continuous at 0
 3) g is a constant function
 4) f is a differentiable function
4. Let $f(x) = \int_0^x |x| dx$, for $-1 \leq x \leq 2$ Where $\int$ represents the integral part function then
 1) $f(x)$ is discontinuous at $x=0$
 2) $f(x)$ is differentiable at $x=1$
 3) $f(x)$ is not differentiable at $x=2$
 4) $f(x)$ is differentiable at $x=2$
5. The function $f(x) = |x^3|$ is
 1) differentiable everywhere
 2) continuous but not differentiable at $x=0$
 3) not a continuous function
 4) a function with range $(0, \infty)$
6. Which of the following function is differentiable at $x=0$
 1) $\cos(|x|) + |x|$ 2) $\cos(|x|) - |x|$
 3) $\sin(|x|) + |x|$ 4) $\sin(|x|) - |x|$
7. The set of all points differentiability of the function $f(x) = \frac{\sqrt{x+1} - x}{\sqrt{x}}$ for $x \neq 0$ and $f(0)=0$ is
 1) $(-\infty, \infty)$ 2) $[0, \infty)$ 3) $(0, \infty)$ 4) $(-\infty, \infty) \setminus \{0\}$
8. For what triplets of real numbers (a, b, c) with $a \neq 0$ the function $f(x)$

$$= \begin{cases} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{cases}$$
 is differentiable for all real x
 1) $\{(a, 1-2a, a) / a \in \mathbb{R}, a \neq 0\}$
 2) $\{(a, 1-2a, c) / a, c \in \mathbb{R}, a \neq 0\}$
 3) $\{(a, b, c) / a, b, c \in \mathbb{R}, a+b+c=1\}$
 4) $\{(a, 1-2a, 0) / a \in \mathbb{R}, a \neq 1\}$
9. Let $f(x) = \frac{|x|(3e^{|x|} + 4)}{2 - e^{|x|}}$, $x \neq 0$ and $f(0)=0$ then
 1) f is not continuous
 2) f is continuous but not differentiable at $x=0$
 3) $f'(0)$ exist 4) $f'(0+) = 2$
10. Given that $f(x)$ is a differentiable function of x and that $f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2$ and that $f(2)=5$. Then $f'(3)$ is equal to
 1) 6 2) 24 3) 15 4) 19
11. If $f(x+y+z) = f(x) \cdot f(y) \cdot f(z)$ for all x, y, z and $f(2)=5$, $f'(0)=3$, then $f'(2)$ equals
 1) 15 2) 9 3) 16 4) 6
12. Suppose that f is a differentiable function with the property that $f(x+y) = f(x) + f(y) + xy$ and $\lim_{h \rightarrow 0} \frac{1}{h} f(h) = 3$, then
 1) f is a linear function 2) $f(x) = 3x + x^2$
 3) $f(x) = 3x + x^2/2$ 4) $f(x) = 3x - x^2/2$
13. if $f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2} \quad \forall x, y \in \mathbb{R}$
 and $f'(0) = -1$, $f(0) = 1$, then $f(2) =$
 1) $\frac{1}{2}$ 2) $-\frac{1}{2}$ 3) 1 4) -1
14. Let $f(x)$ be differentiable function such that $f\left(\frac{x+y}{1-xy}\right) = f(x) + f(y) \quad \forall x$ and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = \frac{1}{3}$ then $f'(1)$ equals
 1) $\frac{1}{4}$ 2) $\frac{1}{6}$ 3) $\frac{1}{12}$ 4) $\frac{1}{8}$
15. Let $f(x)$ be defined for all $x > 0$ and be continuous. Let $f(x)$ satisfy $f\left(\frac{x}{y}\right) = f(x) - f(y)$ for all x, y and $f(e)=1$ Then which of the following may be true
 1) $f(x)$ is bounded 2) $f\left(\frac{1}{x}\right) \rightarrow 0$ as $x \rightarrow \infty$
 3) $x f(x) \rightarrow 1$ as $x \rightarrow 0+$ 4) $f'(x) = \frac{1}{x}$
16. Let $f(x) = \begin{cases} A + \sin^{-1}(x+B), & \forall x \geq 1 \\ x, & \forall x < 1 \end{cases}$ is differentiable then
 1) $A=-1, B=-1$ 2) $A=1, B=-1$
 3) $A=B=1$ 4) $A=0, B=1$
17. Let $f(x) = \begin{cases} x^2 - 1, & x \leq 1 \\ k(x-1), & x > 1 \end{cases}$ then
 1) f is continuous for only finitely many values of k
 2) f is discontinuous at $x=1$
 3) f is differentiable only when $k=2$
 4) there are infinitely many values of k for which f is differentiable

18. Let $f(x) = \begin{cases} x^n \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then $f(x)$ is continuous

but not differentiable at $x=0$ if

- 1) $n \in (0, 1]$ 2) $n \in [1, \infty)$
3) $n \in (-\infty, 0)$ 4) $n=0$

19. Let $f(x) = \begin{cases} \int_0^x \{1 + |1-t|\} dt & \text{if } x > 2 \\ 5x-7 & \text{if } x \leq 2 \end{cases}$, then

- 1) f is not continuous at $x=2$
2) f is continuous but not differentiable at $x=2$
3) f is differentiable everywhere
4) $f'(2+)$ does not exist

20. The values of a and b such that the function f

defined as $f(x) = \begin{cases} ax^2 - b, & |x| < 1 \\ -\frac{1}{|x|}, & |x| \geq 1 \end{cases}$ is

differentiable are

- 1) $a=1, b=-1$ 2) $a=\frac{1}{2}, b=\frac{1}{2}$
3) $a=\frac{1}{2}, b=\frac{3}{2}$ 4) $a=\frac{3}{2}, b=\frac{3}{2}$

21. Let the function $f(x) = \begin{cases} \frac{\sin^3 x^2}{x} & \forall x \neq 0 \\ 0 & \forall x = 0 \end{cases}$

- 1) is continuous but not derivable at $x=0$
2) neither continuous nor differentiable at $x=0$
3) continuous but not differentiable at $x=1$
4) continuous and differentiable at $x=0$

22. The function $f(x) = \begin{cases} x^4 + x^2 - x + 2 & \text{for } x \leq 1 \\ 3x^3 - x^2 + x & \text{for } x > 1 \end{cases}$ is

- 1) continuous everywhere
2) differentiable everywhere
3) differentiable at $x=1$
4) such that f' exists everywhere but f' is not continuous at $x=1$

23. Let $f(x)$ be defined by $f(x) = \begin{cases} \sin 2x & \text{if } 0 < x \leq \frac{\pi}{6} \\ ax+b & \text{if } \frac{\pi}{6} < x \leq 1 \end{cases}$

The values of a and b such that f and f' are continuous, are

- 1) $a=1, b=\frac{1}{\sqrt{2}} + \frac{\pi}{6}$ 2) $a=\frac{1}{\sqrt{2}}, b=\frac{1}{\sqrt{2}}$
3) $a=1, b=\frac{\sqrt{3}}{2} - \frac{\pi}{6}$ 4) $a=\frac{\sqrt{3}}{2}, b=\frac{\sqrt{3}}{2} + \frac{\pi}{6}$

24. $f(x) = \begin{cases} b \sin^{-1}\left(\frac{x+c}{2}\right), & -\frac{1}{2} < x < 0 \\ \frac{1}{2} & \text{at } x=0 \\ \frac{e^{\frac{ax}{2}} - 1}{x} & 0 < x < \frac{1}{2} \end{cases}$

If $f(x)$ is differentiable at $x=0$ and $|c| < \frac{1}{2}$ then

- 1) $a=1$ and $64b^2 + c^2 = 4$
2) $a=0$ and $64b^2 + c^2 = 2$
3) $a=2$ and $64b^2 + c^2 = 1$
4) $a=3$ and $64b^2 + c^2 = 3$

25. The function f defined by

$f(x) = \begin{cases} \frac{\sin x^2}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ is

- 1) continuous and derivable at $x=0$
2) neither continuous nor derivable at $x=0$
3) continuous but not derivable at $x=0$
4) continuous but not derivable at $x=1$

26. If $f(x) = \begin{cases} \frac{|x+2|}{\tan^{-1}(x+2)}, & x \neq -2 \\ 2, & x = -2 \end{cases}$, then $f(x)$ is

- 1) continuous at $x=-2$
2) not continuous at $x=-2$
3) differentiable at $x=-2$
4) continuous but not derivable at $x=-2$

27. If a function $f(x)$ is defined as $f(x) = \begin{cases} \frac{x}{\sqrt{x^2}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

then

- 1) $f(x)$ is continuous at $x=0$ but not differentiable at $x=0$
2) $f(x)$ is continuous as well as differentiable at $x=0$
3) $f(x)$ is not continuous at $x=0$
4) $f(x)$ is continuous everywhere

28. At the point $x=1$, the function

$f(x) = \begin{cases} x^3 - 1; & \text{for } 1 < x < \infty \\ x - 1; & \text{for } -\infty < x \leq 1 \end{cases}$ is

- 1) continuous and differentiable
2) continuous and not differentiable
3) discontinuous and differentiable
4) discontinuous and not differentiable

29. Let $f(x) = \begin{cases} (x-1)\sin\frac{1}{x-1}, & \text{if } x \neq 1 \\ 0, & \text{if } x = 1 \end{cases}$

Then which one of the following is true?

[AIEEE 2008]

- 1) f is neither differentiable at $x = 0$ nor at $x = 1$
- 2) f is differentiable at $x = 0$ and at $x = 1$
- 3) f is differentiable at $x = 0$ but not at $x = 1$
- 4) f is differentiable at $x = 1$ not at $x = 0$

30. If $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ exists finitely, then

- 1) $\lim_{x \rightarrow c} f(x) = f(c)$ 2) $\lim_{x \rightarrow c} f'(x) = f'(c)$
- 3) $\lim_{x \rightarrow c} f(x)$ does not exist
- 4) $\lim_{x \rightarrow c} f(x)$ may or may not exist

31. Let $f(x)$ be a function differentiable at $x=c$.

Then $\lim_{x \rightarrow c} f(x)$ equals

- 1) $f'(c)$ 2) $f(c)$ 3) $\frac{1}{f'(c)}$ 4) $f(c)$

32. If $x+4|y|=6y$, then y as a function of x is

- 1) continuous at $x=0$ 2) derivable at $x=0$
- 3) $\frac{dy}{dx} = \frac{1}{2}$ for all x 4) $\frac{dy}{dx} = 0$ for all x

33. The set onto which the derivative of the function $f(x) = x(\log x - 1)$ maps the set $[1, \infty)$ is

- 1) $[1, \infty)$ 2) $(0, \infty)$ 3) $[0, \infty)$ 4) $(0, 0)$

34. Let $g(x) = \begin{cases} e^{2x} & \forall x < 0 \\ e^{-2x} & \forall x \geq 0 \end{cases}$ then $g(x)$ does not satisfies the condition

- 1) continuous $\forall x \in \mathbb{R}$
- 2) not differentiable at $x=0$
- 3) continuous $\forall x \in \mathbb{R}$ & non differentiable at $x=0$
- 4) $g(x)$ is continuous & differentiable everywhere

35. If the function $f(x) = \left[\frac{(x-5)^3}{A} \right] \sin(x-5)$

+ $\cos(x-2)$, where $[.]$ denotes the greatest integer function and $a \in \mathbb{R}$, is continuous and differentiable in $(7, 9)$ then

- 1) $A \in [8, 64]$ 2) $A \in (0, 8]$
- 3) $A \in [64, \infty)$ 4) $A \in (0, 0)$

36. Let $f(x) = [x]^2 + \sqrt{\{x\}}$, where $[.]$ & $\{\}$ respectively denotes the greatest integer and fractional part of functions, then

- 1) $f(x)$ is continuous at all integral points
- 2) $f(x)$ is not differentiable $\forall x \in \mathbb{I}$
- 3) $f(x)$ is discontinuous as $x \in \mathbb{I} - \{1\}$
- 4) $f(x)$ is continuous & differentiable at $x=0$

37. If $f(x) = \sqrt{x+2\sqrt{2x-4}} + \sqrt{x-2\sqrt{2x-4}}$ then $f(x)$ is differentiable on

- 1) $(-\infty, \infty)$ 2) $[2, \infty) - \{4\}$ 3) $[2, \infty)$ 4) $(0, \infty)$

38. Let $f(x) = (x+|x|)|x|$, then for all x

- 1) f is continuous for some x
- 2) f is differentiable for some x
- 3) f' is continuous 4) f'' is continuous

39. If $f(x) = x(\sqrt{x} - \sqrt{x+1})$ then

- 1) f is continuous but not differentiable at $x=0$
- 2) f is differentiable at $x=0$
- 3) f is differentiable but not continuous at $x=0$
- 4) f is not differentiable at $x=0$

40. The number of points at which the function $f(x) = |x-0.5| + |x-1| + \tan x$ does not have a derivative in the interval $(0, 2)$ is

- 1) 1 2) 2 3) 3 4) 4

41. Suppose $f(x)$ is differentiable at $x=1$

and $\lim_{h \rightarrow 0} \frac{1}{h} f(1+h) = 5$, then $f'(1)$ equals

- 1) 3 2) 4 3) 5 4) 6

42. The set of points where $f(x) = \frac{x}{1+|x|}$ is differentiable is

- 1) $(-\infty, 0) \cup (0, \infty)$ 2) $(-\infty, -1) \cup (-1, \infty)$
- 3) $(-\infty, \infty)$ 4) $(0, \infty)$

43. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$f(x) = \min\{x+1, |x|+1\}$, Then which of the following is true? [AIEEE - 2007]

- 1) $f(x)$ is differentiable everywhere
- 2) $f(x)$ is not differentiable at $x=0$
- 3) $f(x) \geq 1$ for all $x \in \mathbb{R}$
- 4) $f(x)$ is not differentiable at $x=1$

44. If function $f(x)$ is differentiable at

$x=a$, then $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x-a}$ is : [AIEEE-2011]

- 1) $-a^2 f'(a)$ 2) $a f(a) - a^2 f'(a)$
- 3) $2a f(a) - a^2 f'(a)$ 4) $2a f(a) + a^2 f'(a)$

45. If $f: (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let

$g(x) = [f(2f(x) + 2)]^2$, then $g'(0)$ is equal to (AIEEE-2010)

- 1) 4 2) -4 3) 0 4) -2

LEVEL-III - KEY

- 1) 4 2) 2 3) 2 4) 3 5) 1 6) 4
 7) 4 8) 1 9) 2 10) 1 11) 1 12) 3
 13) 4 14) 2 15) 4 16) 2 17) 3 18) 1
 19) 2 20) 3 21) 4 22) 1 23) 3 24) 1
 25) 1 26) 2 27) 3 28) 2 29) 3 30) 1
 31) 4 32) 1 33) 3 34) 3 35) 3 36) 3
 37) 2 38) 3 39) 2 40) 3 41) 3 42) 3
 43) 1 44) 3 45) 2

LEVEL-III - HINTS

- $$f(x) = \begin{cases} \cos x - \sin x & \text{for } x \in (0, \pi/4) \\ \sin x - \cos x & \text{for } x \in (\pi/4, \pi/2) \end{cases}$$
- Let $f(x) = ax^2 + bx + c$. As $f(x) > 0$ for all $x \in \mathbb{R}$, we must have, $a > 0$ and $b^2 - 4ac < 0$

$$g(x) = ax^2 + bx + c + (2ax + b) + 2a + 0 + (x^2) \cdot 0$$

$$= ax^2 + (b+2a)x + b + c + 2a$$

Discriminant of $g(x) = (b+2a)^2 - 4a(b+c+2a)$

$$= -4a^2 + (b^2 - 4ac) < 0$$

Thus $g(x) > 0$ for $x \in \mathbb{R}$
- By options (2) is correct.
- $$f(x) = \begin{cases} -x, & -1 \leq x < 0 \\ 0, & 0 \leq x < 1 \\ x, & 1 \leq x < 2 \\ 4, & 2 \leq x < 3 \text{ or } x = 2 \text{ (as given)} \end{cases}$$

Now $\lim_{x \rightarrow 0^-} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = 0$ then $f(0) = 0$
 $\therefore f(x)$ is continuous at $x=0$

Again $\lim_{x \rightarrow 1^-} f(x) = 0$, $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1} x = 1$
 $f(x)$ is discontinuous at $x=1$ so we say $f(x)$ is non-differentiable at $x=1$

Again $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2} x = 2 \neq f(x)_{x=2} \text{ i.e. } 4$
 $\therefore f(x)$ is discontinuous at $x=2$
 So non-differentiable at $x=2$
- $$f(x) = |x^3| \quad \text{then } f(x) = \begin{cases} -x^3 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ x^3 & \text{if } x > 0 \end{cases}$$

$f'(0^-) = 0$; $f'(0^+) = 0 \Rightarrow f'(0^-) = f'(0^+)$
 then f is differentiable at $x=0$
 i.e f is continuous at $x=0$
- $\cos |x| = \cos x$ is differentiable at $x=0$, but $|x|$ is not differentiable at $x=0$, Hence 1 & 2 options are not correct.

$$f(x) = \sin |x| - |x| = \begin{cases} -\sin x + x & x < 0 \\ \sin x - x & x \geq 0 \end{cases}$$

is differentiable at $x=0$

7. $f'(x)$ is not exist at $x=0$

$\therefore f(x)$ is differentiable $\forall x \in (-\infty, \infty) - \{0\}$

$$8. \quad f'(x) = \begin{cases} 1 & x \leq 1 \\ 2ax + b & x > 1 \end{cases}$$

f is differentiable, then $f'(1^+) = f'(1^-)$

$$\Rightarrow 1 = 2a + b \Rightarrow b = 1 - 2a \text{ --- (1)}$$

Given $f(x)$ is continuous at $x=1$ then $1 = a + b + c$

$$9. \quad f'(0^-) = \lim_{x \rightarrow 0^-} \frac{\frac{|x|(3e^{\frac{1}{x}} + 4)}{2 - e^{\frac{1}{x}}} - 0}{x - 0} = \lim_{x \rightarrow 0^-} \frac{-x(3e^{-\frac{1}{x}} + 4)}{2 - e^{-\frac{1}{x}}} - 0$$

$$= \frac{3e^{-\infty} + 4}{2 - e^{-\infty}} = 2$$

$$f'(0^+) = \lim_{x \rightarrow 0^+} \frac{\frac{|x|(3e^{\frac{1}{x}} + 4)}{2 - e^{\frac{1}{x}}} - 0}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x(3e^{\frac{1}{x}} + 4)}{2 - e^{\frac{1}{x}}} - 0$$

$$= \lim_{x \rightarrow 0^+} \frac{\left(3 + \frac{4}{e^{\frac{1}{x}}}\right)}{\frac{2}{e^{\frac{1}{x}}} - 1} = \frac{3}{-1} = -3 \Rightarrow f'(0^-) \neq f'(0^+)$$

then f is not differentiable at $x=0$ and continuous at $x=0$

10. We have, $f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + f(1) - 2$$

$$\Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

(since $f(1) = 2$ putting $x=y=1$)

$$\Rightarrow f(x) = x^n + 1 \Rightarrow f(2) = x^2 + 1 \text{ (since } f(2)=5)$$

$$\Rightarrow n = 2$$

$$\therefore f(x) = x^2 + 1 \Rightarrow f(3) = 10$$

11. We have, $f(x+y+z) = f(x)f(y)f(z)$ for all x, y, z

$$\Rightarrow f(0) = f(0)f(0)f(0) \text{ (putting } x=y=z=0)$$

$$\Rightarrow f(0)\{1 - (f(0))^2\} = 0 \Rightarrow f(0) = 1$$

($\because f(0) = 0 \Rightarrow f(x) = 0$ for all x)

Putting $z=0$ and $y=2$, we get

$$f(x+2) = f(x)f(2)f(0) \Rightarrow f(x+2) = 5f(x) \text{ for all } x$$

$$\Rightarrow f'(2) = 5f'(0) = 5 \times 3 = 15$$

hence f is differentiable everywhere

$$12. \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) + f(h) + xh - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} (h) + x = 3 + x$$

Hence $f(x) = 3x + \frac{x^2}{2} + c$. Putting $x=y=0$ in the given equation, we have $f(0) = f(0) + f(0) + 0 \Rightarrow f(0) = 0$.

Thus $c=0$ and $f(x) = 3x + \frac{x^2}{2}$

13. Take $f(x) = ax + b$

$$14. f\left(\frac{x+y}{1-xy}\right) = f(x) + f(y) \Rightarrow f(x) = A \tan^{-1} x$$

$$\text{Now, } \lim_{x \rightarrow 0} \frac{f(x)}{x} = A \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x}, \therefore A = \frac{1}{3} \therefore f(x) = \frac{1}{3} \tan^{-1} x$$

15. A function satisfying the given functional equation is $f(x) = \log x$. Clearly $f(x)$ is not bounded and $f(\frac{1}{x}) = -\log x \not\rightarrow 0$ as $x \rightarrow \infty$.

Also $x f(x) = x \log x \not\rightarrow 1$ as $x \rightarrow 0$. But $f'(x) = \frac{1}{x}$.

In fact, it can be shown that any continuous functions satisfying the given functional equation is of the form $k \log x$, for some $k > 0$.

Since $f(e) = 1$ so $k=1$

16. As $x \geq 1$ $\sin^{-1}(x+B)$ is defined when $B = -1$ from the options. f is differentiable $\Rightarrow f$ is continuous.

$$17. f(x) = \begin{cases} x^2 - 1 & \text{if } x \leq 1 \\ k(x-1) & \text{if } x > 1 \end{cases} \text{ then } f'(x) = \begin{cases} 2x & \text{if } x \leq 1 \\ k & \text{if } x > 1 \end{cases}$$

$$f'(1-) = f'(1+) \text{ i.e. } 2(1) = k \text{ then } k=2$$

18. since $f(x)$ is continuous at $x=0$, therefore

$$\lim_{x \rightarrow 0} f(x) = f(0) = 0 \Rightarrow \lim_{x \rightarrow 0} x^n \sin\left(\frac{1}{x}\right) = 0 \Rightarrow n > 0$$

$f(x)$ is differentiable at $x=0$ if

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \text{ exists finitely}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^n \sin\left(\frac{1}{x}\right) - 0}{x} \text{ exists finitely}$$

$$\Rightarrow \lim_{x \rightarrow 0} x^{n-1} \sin\left(\frac{1}{x}\right) \text{ exists finitely } \Rightarrow n-1 > 0 \Rightarrow n > 1$$

If $n \leq 1$, then $\lim_{x \rightarrow 0} x^{n-1} \sin\left(\frac{1}{x}\right)$ does not exist and hence

$f(x)$ is not differentiable at $x=0$

hence $f(x)$ is continuous but not differentiable at $x=0$ for $0 < n \leq 1$, i.e. $n \in (0, 1]$

$$19. \text{ For } x > 2, \int_0^x \{1 + |1-t|\} dt = \int_0^1 \{1 + |1-t|\} dt + \int_1^x \{1 + |1-t|\} dt$$

$$= \int_0^1 (2-t) dt + \int_1^x t dt = 1 + \frac{x^2}{2}$$

$$\text{Thus, } f(x) = \begin{cases} 1 + \frac{x^2}{2}, & x > 2 \\ 5x - 7, & x \leq 2 \end{cases}$$

$$\lim_{x \rightarrow 2^+} f(x) = 1 + \frac{2^2}{2} = 3 = f(2) = \lim_{x \rightarrow 2^-} f(x)$$

$$f'(2+) = \lim_{h \rightarrow 0^+} \frac{1 + \frac{(2+h)^2}{2} - 3}{h} = \lim_{h \rightarrow 0^+} \frac{\frac{1}{2}h^2 + 2h}{h} = 2$$

$$f'(2-) = \lim_{h \rightarrow 0^+} \frac{5(2+h) - 7 - 3}{h} = 5$$

Hence f is continuous but not differentiable at $x=2$

20. Since every differentiable function is continuous, so we must have

$$\lim_{x \rightarrow 1^-} f(x) = f(1) \Rightarrow a - b = -1$$

for f to be differentiable, $f'(1-) = f'(1+)$

$$\Rightarrow \lim_{h \rightarrow 0^-} \left[\frac{a(1+h)^2 - b + 1}{h} \right] = \lim_{h \rightarrow 0^-} \left[\frac{-1|1+h| + 1}{h} \right]$$

$$= \lim_{h \rightarrow 0^-} \left[\frac{a(2h+h^2)}{h} \right] = \lim_{h \rightarrow 0^-} \frac{h}{h(1+h)} \text{ (as } a-b=-1)$$

$$\Rightarrow 2a = 1, \text{ Hence } a = \frac{1}{2} \text{ and } b = \frac{3}{2}$$

$$21. \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} (x \sin^2 x^2) = \lim_{x \rightarrow 0} \left(\frac{\sin x^2}{x^2} \right) \cdot \lim_{x \rightarrow 0} (x \sin^2 x^2) = 0 = f(0)$$

$$\text{and LHD at } x=0 \text{ is } \lim_{h \rightarrow 0} \frac{f(0+h) - 0}{h} = \frac{\sin^3 h^2 - 0}{h} = 0$$

RHD=0

$\therefore f(x)$ is differentiable at $x=0$

$$22. f(x) = \begin{cases} x^4 + x^2 - x + 2 & \text{for } x \leq 1 \\ 3x^3 - x^2 + x & \text{for } x > 1 \end{cases}$$

$$f'(x) = \begin{cases} 4x^3 + 2x - 1 & \text{for } x \leq 1 \\ 9x^2 - 2x + 1 & \text{for } x > 1 \end{cases}$$

$$f'(1-) = 5, f'(1+) = 8 \Rightarrow f'(1-) \neq f'(1+)$$

$\therefore f(x)$ is not differentiable at $x=1$

$$\lim_{x \rightarrow 1} f(x) = f(1) \text{ satisfied, then } f \text{ is continuous at } x=1$$

$\therefore f(x)$ is continuous at everywhere

$$23. f(x) = \begin{cases} \sin 2x & \text{if } 0 < x \leq \frac{\pi}{6} \\ ax + b & \text{if } \frac{\pi}{6} < x \leq 1 \end{cases}$$

$$\text{then } f'(x) = \begin{cases} 2\cos 2x & \text{if } 0 < x \leq \frac{\pi}{6} \\ a & \text{if } \frac{\pi}{6} < x \leq 1 \end{cases}$$

f and f' are continuous

$$\lim_{x \rightarrow \frac{\pi}{6}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{6}^+} f(x) \Rightarrow \lim_{x \rightarrow \frac{\pi}{6}^-} \sin 2x = \lim_{x \rightarrow \frac{\pi}{6}^+} ax + b$$

$$\frac{\sqrt{3}}{2} = \frac{a\pi}{6} + b \Rightarrow b = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

$$\lim_{x \rightarrow \frac{\pi}{6}^-} f'(x) = \lim_{x \rightarrow \frac{\pi}{6}^+} f'(x) \text{ then } \lim_{x \rightarrow \frac{\pi}{6}^-} 2\cos 2x = \lim_{x \rightarrow \frac{\pi}{6}^+} a$$

i.e. $a=1$

$$24. \text{ Find } f'(x) \text{ and } f'(0^+) = f'(0^-)$$

$$25. f(0) = \lim_{x \rightarrow 0} \frac{\sin x^2 - 0}{x - 0} = 1$$

then f is differentiable at $x=0$

$$26. \lim_{x \rightarrow 2} f(x) = \lim_{h \rightarrow 0} f(2-h) = \lim_{h \rightarrow 0} \frac{-2-h+2}{\tan^{-1}(-2-h+2)}$$

$$= \lim_{h \rightarrow 0} \frac{h}{\tan^{-1}(-h)} = \lim_{h \rightarrow 0} \frac{-h}{\tan^{-1}(h)} = -1$$

$$\text{and } \lim_{x \rightarrow 2} f(x) = \lim_{h \rightarrow 0} f(-2+h) = \lim_{h \rightarrow 0} \frac{-2+h+2}{\tan^{-1}(-2+h+2)}$$

$$= \lim_{h \rightarrow 0} \frac{h}{\tan^{-1}(h)} = 1 \quad \therefore \lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

so, f is neither continuous nor differentiable at $x=2$

$$27. \text{ we have, } f(x) = \begin{cases} \frac{x}{\sqrt{x^2}}, & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \\ 0, & x = 0 \end{cases}$$

$f(x)$ is not continuous at $x=0$

$$28. \text{ we have, } \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x-1) = 0 \text{ and}$$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x^3 - 1) = 0. \text{ Also } f(1) = 1 - 1 = 0$$

so, $f(x)$ is continuous at $x=1$ Clearly $Lf'(1) = 2$ and

$Rf'(1) = 3$. Therefore, $f(x)$ is not differentiable at $x=1$

$$29. Lf'(1) = \lim_{h \rightarrow 0} \frac{(1-h-1)\sin\left(\frac{1}{1-h-1}\right) - 0}{-h}$$

$$= -\lim_{h \rightarrow 0} \sin \frac{1}{h} \text{ similarly } Rf'(1) = \lim_{h \rightarrow 0} \sin \frac{1}{h}$$

' f ' is not differentiable at $x=1$, Clearly ' f ' is differentiable at $x=0$.

$$\text{As } Lf'(0) = Rf'(0) = \cos 1 - \sin 1$$

30. since $f(x)$ is differentiable at $x=c$, therefore it is continuous at $x=c$. Hence $\lim_{x \rightarrow c} f(x) = f(c)$

31. since $f(x)$ is differentiable at $x=c$, therefore it is continuous at $x=c$. Hence $\lim_{x \rightarrow c} f(x) = f(c)$

32. We have, $x+4|y|=6y$

$$\Rightarrow \begin{cases} x-4y=6y, & \text{if } y < 0 \\ x+4y=6y, & \text{if } y \geq 0 \end{cases}$$

$$\Rightarrow y = \begin{cases} \frac{1}{2}x, & \text{if } x \geq 0 \\ \frac{1}{10}x, & \text{if } x < 0 \end{cases} \Rightarrow y = f(x) = \begin{cases} \frac{1}{2}x, & \text{if } x \geq 0 \\ \frac{1}{10}x, & \text{if } x < 0 \end{cases}$$

clearly, $y = f(x)$ is continuous at $x=0$ but it is not differentiable at $x=0$

$$33. f'(x) = \log x - 1 + x \left(\frac{1}{x} \right) = \log x$$

since $\log x$ is increasing function so f' maps $[1, \infty)$ onto $[0, \infty)$

$$34. \text{ Given } g(x) = \begin{cases} e^{2x} & \forall x < 0 \\ e^{-2x} & \forall x \geq 0 \end{cases}, \text{ then}$$

$$g'(x) = \begin{cases} 2e^{2x} & \forall x < 0 \\ -2e^{-2x} & \forall x \geq 0 \end{cases}$$

$\therefore$ LHD at $x=0$ is $2e^0=2$

RHD at $x=0$ is $-2e^0=-2$

As LHD at $x=0 \neq$ RHD at $x=0$

$\therefore g(x)$ is not differentiable at $x=0$

Again $\lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^-} e^{2x} = 1$ and $\lim_{x \rightarrow 0^+} g(x) = \lim_{x \rightarrow 0^+} e^{-2x} = 1$

$$\therefore \lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^+} g(x)$$

$\therefore g(x)$ is continuous at $x=0$, more over $g(x)$ is continuous $\forall x \in \mathbb{R}$

35. $[x]$ is not continuous and differentiable at integral values (points)

So $f(x)$ is continuous and differentiable in $(7,9)$

$$\text{if } \left[\frac{(x-5)^3}{A} \right] = 0 \Rightarrow A \geq (9-5)^3 \Rightarrow A \geq 64 \therefore A \in [64, \infty)$$

36. If $k \in I$

$$\lim_{x \rightarrow k^+} f(x) = k^2 - 0 \Rightarrow \lim_{x \rightarrow k^-} f(x) = (k-1)^2 + 1$$

Again $\lim_{x \rightarrow k^-} f(x) = \lim_{x \rightarrow k^+} f(x) = f(k)$

$$k^2 = (k-1)^2 + 1 \Rightarrow 2k = 2 \text{ then } k = 1$$

i.e., $f(x)$ is continuous at $k=1$ and no other integral point.

So $f(x)$ is discontinuous for all integral points except $x=1$

$$37. f(x) = \sqrt{\left(\frac{\sqrt{2x-y}}{2}\right)^2} + 2 + \sqrt{2x-4}$$

$$+ \sqrt{\frac{(2x-4)^2}{2}} + 2 - 2\sqrt{2x-4}$$

$$\Rightarrow f(x) = \frac{1}{\sqrt{2}} \left[(\sqrt{2x-4} + 2) + |\sqrt{2x-4} - 2| \right]$$

$$\Rightarrow \left[1 + \frac{dy}{dx} \right] = 0 \quad \therefore \frac{dy}{dx} = -1$$

$$\Rightarrow f(x) = \begin{cases} \frac{4}{\sqrt{2}} & \text{if } \sqrt{2x-4} - 2 < 0 \text{ i.e. } \sqrt{2x-4} < 2 \text{ i.e. } x < 4 \\ \sqrt{x-2} & \text{if } \sqrt{2x-4} \geq 2 \text{ i.e. } 2x \geq 8 \text{ i.e. } x \geq 4 \\ 2\sqrt{2} & \text{if } 2 \leq x < 4 \end{cases}$$

$$\text{Now } f'(x) = \frac{1}{2\sqrt{x-2}} \text{ if } 4 \leq x < \infty$$

$$0 \text{ if } 2 \leq x \leq 4$$

But when $x=4$, $f'(4) = \frac{1}{0}$ not defined

$\therefore f(x)$ is differentiable on $[2,4) \cup (4, \infty)$

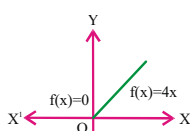
i.e. $x \in [2, \infty) - \{4\}$

$$38. \text{ We have, } f(x) = \begin{cases} (x-x)(-x) = 0, & x < 0 \\ (x+x)x = 2x^2, & x \geq 0 \end{cases}$$

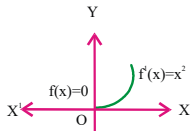
As is evident from the graph of $f(x)$ that it is continuous and differentiable for all x , also

$$f'(x) = \begin{cases} 0, & x < 0 \\ 4x, & x \geq 0 \end{cases}$$

clearly, $f(x)$ continuous for all x but it is not differentiable at $x=0$



fig(i)



fig(ii)

39. we have $f(x) = x(\sqrt{x} - \sqrt{x+1})$

$$\Rightarrow f'(x) = (\sqrt{x} - \sqrt{x+1}) + \frac{x(\sqrt{x+1} - \sqrt{x})}{2\sqrt{x}\sqrt{x+1}} \text{ for } x \neq 0, -1.$$

Also $f'(0+) = -1$, so $f(x)$ is differentiable and hence continuous at $x=0$

40. we have, $f(x) = |x - 0.5| + |x - 1| + \tan x, 0 < x < 2$

$$= \begin{cases} -2x + 1.5 + \tan x, & 0 < x < 0.5 \\ 0.5 + \tan x, & 0.5 \leq x < 1 \\ 2x - 1.5 + \tan x, & 1 \leq x < 2 \end{cases}$$

It is evident from the above definition that $Lf'(0.5) \neq Rf'(0.5)$ and $Lf'(1) \neq Rf'(1)$. Also, the function is not continuous at $x = \frac{\pi}{2}$. So, it cannot be differentiable at three points

$$41. f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$\text{given that } \lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5 \text{ and hence}$$

$$f(1) = 0$$

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 5$$

$$42. f'(x) = \begin{cases} \frac{x}{(1-x)^2}; & x < 0 \\ \frac{x}{(1+x)^2}; & x \geq 0 \end{cases}$$

$$43. f(x) = x+1, \forall x \in \mathbb{R}$$

$$44. \lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$$

$$\lim_{x \rightarrow a} \frac{2xf(a) - a^2 f'(x)}{1}$$

$$= 2af(a) - a^2 f'(a)$$

$$45. g'(x) = 2[f(2f(x)+2)] \times f'((2f(x))+2) \times 2f'(x)$$

$$g'(0) = [f(2f(0)+2)] \times f'((2f(0))+2) \times 2f'(0)$$

$$= 2f(0) \times f'(0) \times 2f'(0) = 2(-1) \times 1 \times 2 \times 1 = -4$$



LEVEL-IV

Here each question contains two statements: Statement -1 (Assertion) and Statement-2 (Reason):

each question has 4 choices (1), (2), (3) and (4) out of which, only one is correct

1) Statement-1 is true, statement -2 is true. Statement-2 is a correct explanation for statement-1

2) Statement-1 is true, statement -2 is true. Statement-2 is not a correct explanation for statement-1

3) Statement-1 is true, statement -2 is false.

4) Statement-1 is false, statement -2 is true.

1. **Statement-1:** if $y = f(x) = \frac{x}{1+|x|}, x \in \mathbb{R}$, then $f(x)$ is differentiable everywhere.

Statement-2: If $f(x) = \frac{x}{1+|x|}, x \in \mathbb{R}$, then

$$f'(x) = \begin{cases} \frac{1}{(1+x)^2}, & x \geq 0 \\ \frac{1}{(1-x)^2}, & x \leq 0 \end{cases}$$

2. **Statement-1:** If $f(x) = [\tan x], x \in \left[0, \frac{\pi}{3}\right]$,

then $f'\left(\frac{\pi}{4}\right)$ does not exist.

Statement-2: $Lf'\left(\frac{\pi}{4}\right) \rightarrow \infty$

3. **Statement-1:** $f(x) = x^n \sin\left(\frac{1}{x}\right)$ is differentiable for all real values of x ($n \geq 2$).
- Statement-2:** For $n \geq 2$, right derivative = left derivative (for all real value of x)
4. **Statement-1:** The function $y = \sin^{-1}(\cos x)$ is not differentiable at the points where $\sin x = 0$.
- Statement-2:** $\frac{dy}{dx} = \frac{-\sin x}{|\sin x|}$, So the function is not differentiable at the points where $\sin x = 0$
5. **Statement-1:** $f(x) = e^{-|x|}$ is differentiable at $x=0$
- Statement-2:** $f(x) = e^{-|x|}$ is continuous at $x=0$
6. Let $f(x) = x|x|$ and $g(x) = \sin x$
- Statement-1:** $g \circ f$ is differentiable at $x=0$ and its derivative is continuous at that point
- Statement-2:** $g \circ f$ is twice differentiable at $x=0$
7. Consider the function, $f(x) = |x-2| + |x-5|, x \in \mathbb{R}$
- Statement-1:** $f'(4) = 0$
- Statement-2:** f is continuous in $[2, 5]$, differentiable in $(2, 5)$ and $f(2) = f(5)$.

LEVEL-IV - KEY

1) 1 2) 3 3) 1 4) 4 5) 4 6) 3 7) 1

LEVEL-IV - HINTS

1. verify at $x=0$
2.
$$Rf' = \lim_{x \rightarrow \frac{\pi}{4}^+} \frac{f(x) - f\left(\frac{\pi}{4}\right)}{x - \frac{\pi}{4}}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}^+} \frac{(\tan x) - 0}{x - \frac{\pi}{4}} = \frac{1}{0} \rightarrow \infty$$
3. $x^n \sin \frac{1}{x}$ is differentiable when $n > 1$
4. $y = \sin^{-1}(\cos x) = \sin^{-1}\left(\sin\left(\frac{\pi}{2} + x\right)\right)$

$$= \frac{\pi}{2} + x \Rightarrow \frac{dy}{dx} = 1$$

$\therefore$ Statement-1 is false
statement-2 is clearly true

5. **Statement-2:**

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} e^{-(-x)} = \lim_{x \rightarrow 0^-} e^x = e^0 = 1 = f(0)$$

statement-1:

$$Lf'(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0^-} \frac{e^{-(-x)} - 1}{x} = \lim_{x \rightarrow 0^-} \frac{e^x - 1}{x} = 1$$

$$\text{and } Rf'(0) = -1$$

6. Let $h(x) = (g \circ f)(x)$

$$\therefore (g \circ f)(x) = g(f(x)) = \sin(x|x|)$$

$$\therefore h(x) = \begin{cases} -\sin x^2 & \text{if } x < 0 \\ \sin x^2 & \text{if } x > 0 \end{cases}$$

$$\therefore h'(x) = \begin{cases} -2x \sin x^2 & \text{if } x < 0 \\ 2x \cos x^2 & \text{if } x > 0 \end{cases}$$

$$\therefore Rh'(0) = \lim_{x \rightarrow 0^+} \frac{h(x) - h(0)}{x}$$

Now

$$= \lim_{x \rightarrow 0^+} \frac{\sin x^2}{x} = 0$$

$$7. f(x) = |x-2| = \begin{cases} x-2, & x-2 \geq 0 \\ 2-x, & x-2 \leq 0 \end{cases}$$

$$= \begin{cases} x-2, & x \geq 2 \\ 2-x, & x \leq 2 \end{cases}$$

$$\text{Similarly, } f(x) = |x-5| = \begin{cases} x-5, & x \geq 5 \\ 5-x, & x \leq 5 \end{cases}$$

$$\therefore f(x) = |x-2| + |x-5|$$

$$= \{x-2+5-x = 3, 2 \leq x \leq 5\}$$

$$\text{Thus } f(x) = 3, 2 \leq x \leq 5$$

$$f'(x) = 0, 2 < x < 5$$

$$f'(4) = 0, \therefore \text{statement 1 is true.}$$

$$\therefore f(2) = 3 \text{ and } \therefore f(5) = |5-2| + 0 = 3$$

$\therefore$ statement-2 is also true and a correct explanation for statement 1.